

Max Booth

CONTACT

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WORK SUMMARY

Mathematician interested in operator theory and algebraic paradigms in DS/ML as well as applications in energy, finance, & discrete systems.

EDUCATION

Johns Hopkins University , Ph.D. Candidate in Applied Math & Statistics	2024—2026
University of Utah , B.S. in Applied Math <i>Summa Cum Laude</i>	2023

EXPERIENCE

Graduate Teaching Assistant (Applied Mathematics & Statistics) Johns Hopkins University <i>Honors Discrete Mathematics</i>— FA24, SP25, FA25, SP26 <i>Intermediate Probability & Statistics</i>— FA 2025	2024—2026
Research Assistant (Biostatistics) The Johns Hopkins University School of Medicine Advisors: Elana Fertig, Genevieve Stein-O'Brien <i>Sponsored by NIH, NCI</i> <i>Python, C++, MATLAB, PhysiCell: model construction & validation</i>	2022—2024
Undergraduate Teaching Assistant (Mathematics) University of Utah Supervisor: Lisa Penfold <i>Tutor, Foundations of Analysis Tutor</i> <i>Grader: MATH 1050, 1080, 1210, 1220, 1310, 1320, 2200, 2210, 2250, 2270, 3150, 3210, & 3220</i>	2019—2022
Private Tutor (Mathematics) Salt Lake City, UT <i>College Algebra, Calculus, Linear Algebra, ODEs, PDEs, Discrete Mathematics, Foundations of Analysis I, & Foundations of Analysis II</i>	2017—2023

PUBLICATIONS & PREPRINTS

[1] **BIWT: a bioinformatics walkthrough for embedding spatial multiomics in agent-based models for virtual cells**
Daniel R Bergman et int, Max Booth et al. *Bioinformatics* (2025), 41 (11), btaf571: <https://doi.org/10.1093/bioinformatics/btaf571>

[2] **Human interpretable grammar encodes multicellular systems biology models to democratize virtual cell laboratories**
Jeanette A I Johnson et int, Max Booth et al. *Cell* (2025), 188 (4711–4733): <https://doi.org/10.1016/j.cell.2025.06.048>

PROJECTS & CERTIFICATES	Quantum Computer Programming: Hands-On Workshop	2026
	IBM Quantum (Asynchronous) <i>Practical Quantum Computing, Qiskit, MaxCut, QAOA</i>	
	PhysiCell Workshop and Hackathon	2023
	Indiana University Bloomington Project: <i>Random Walks ... with Modulated Responses to Physical Barriers</i>	
	Research Internship— PhysiCell Data Integration & Analysis	2022
	The Johns Hopkins University School of Medicine Supervisors: Elana Fertig, Jeanette A I Johnson	
AWARDS	President's Scholarship	2019—2023
	University of Utah	
CONFERENCE ATTENDANCE	IMAG/MSM Past2Future Meeting	2023